Case 6 — Mr. George Miller

75 years old · Male | Parkinson's Disease · 12 years duration

Target: 70%+ | Neurological Physiotherapy

■ PROBLEM LIST

- Freezing of gait (FOG) — turning-triggered, confirmed by falls context
- Shuffling gait — reduced step length, increased cadence, poor heel clearance
- On-off motor fluctuations — sudden loss of mobility unpredictably
- Severe postural instability (Hoehn & Yahr IV — expected)
- No ROM deficits — impairment is central/neurological, not musculoskeletal
- Likely significant rigidity (advanced disease)
- Probable autonomic dysfunction (to confirm with lying/standing BP)
- Likely dual-task cost elevation (to quantify objectively)
- High falls risk — two falls in past month, both at home
- Unable to safely turn in confined spaces (kitchen)
- Likely housebound or near-housebound — community mobility severely restricted
- Increasing ADL dependence during "off" periods

■ RED FLAGS

None identified

■ PATHOPHYSIOLOGY

- 2 Pathophysiology of Working Diagnosis What Is Happening Neurologically — Advanced PD After 12 years, Mr.
- Miller's disease has progressed to a state of near-total dopaminergic neuron loss in the substantia nigra, with Lewy body pathology (Braak stage V–VI) extensively involving the frontal cortex, limbic system, brainstem reticular formation, and autonomic nervous system.
- This explains why FOG is most dangerous in confined, complex environments.
- Motor fluctuations — the "on-off" phenomenon — arise because after 12 years of levodopa therapy, the remaining dopaminergic neurons have lost their capacity to buffer and gradually release dopamine.
- Dopamine availability in the striatum now mirrors levodopa plasma levels directly, oscillating sharply with each dose cycle.

■ DIFFERENTIAL DIAGNOSES

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■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Freezing of Gait Questionnaire (FOG-Q) + Timed 360° Turn Test

Test 1: Freezing of Gait Questionnaire (FOG-Q) + Timed 360° Turn Test What it measures: FOG-Q is a validated 6-item patient-reported outcome measure quantifying the frequency, duration, and severity of freezing episodes across different contexts (starting to walk, turning, going

Pull Test (Postural Instability — MDS-UPDRS Item 3.12)

Test 2: Pull Test (Postural Instability — MDS-UPDRS Item 3.12) What it measures: Automatic postural righting responses — whether Mr. Miller can recover his balance from an unexpected backward perturbation. At Hoehn & Yahr IV, I expect severely impaired postural reflexes. This tes

Dual-Task Walking Assessment (10-Metre Walk — Single vs. Dual Task)

Test 3: Dual-Task Walking Assessment (10-Metre Walk — Single vs. Dual Task) What it measures: The "Dual-Task Cost" (DTC) — the percentage decrease in gait performance when a concurrent cognitive task is added. In advanced PD, walking is no longer automatic; it requires significan

■ TREATMENT PLAN

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